



Data Science Intern at Data Glacier

Project: Hate Speech Detection using Transformers (Deep Learning)

Week 7: Deliverable

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Table of Contents:

1. Project Plan	3
2. Problem Description	3
3. Business Understanding	3

1. Project Plan

Weeks	Date	plan
Weeks 07	May 19, 2025	Problem Description, Business understanding, Data Intake Report
Weeks 08	May 26, 2025	Data Preprocessing
Weeks 09	June 2, 2025	Data Cleansing and Transformation
Weeks 10	June 9, 2025	EDA performed
Weeks 11	June 16, 2025	EDA Presentation and Proposed Modelling Technique
Weeks 12	June 23, 2025	Model Selection and Model Building
Weeks 13	June 30, 2025	Final Submission (Report + Code + Presentation)

2. Problem Description

Hate speech poses a significant challenge on digital platforms, often manifesting as verbal, written, or behavioral communication targeting individuals or groups based on religion, ethnicity, nationality, race, gender, or other identity traits. With the rise of social media usage, particularly platforms like Twitter, monitoring and mitigating such harmful content has become essential.

This project aims to build an automated hate speech detection system using advanced Natural Language Processing (NLP) techniques and transformer-based deep learning models. The task is framed as a binary text classification problem, where the model learns to distinguish between hateful and non-hateful tweets. The goal is to create a reliable and scalable model capable of supporting content moderation efforts by detecting hate speech in real time.

3. Business Understanding

Social media platforms are essential tools for global communication but also serve as mediums for the spread of hate speech and abusive content. The unchecked proliferation of such content can lead to social unrest, legal issues, and reputational damage for platform providers.

Businesses operating these platforms are increasingly under pressure to ensure user safety and compliance with regulations on digital conduct.

By deploying an automated hate speech detection system, companies can significantly enhance their moderation capabilities. This not only helps maintain a safer online environment but also improves user trust and brand integrity. From a data science perspective, this project addresses key business goals such as minimizing manual moderation efforts, reducing harmful interactions, and supporting ethical AI practices in content governance.